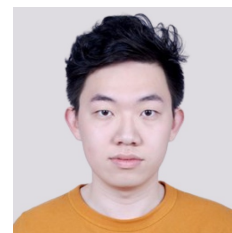


BOYU ZHU

Computer Science PhD Student

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EDUCATION

PhD, Biometrics and Explainability of Artificial Intelligence <i>University of Bordeaux, Talence, France</i>	Dec. 2024 – Present
Master's Degree, Data Science <i>Paris-Saclay University, Orsay, France</i>	Sept. 2020 – Sept. 2023
Bachelor's Degree, Computer Science <i>Paris-Saclay University, Orsay, France</i>	Sept. 2019 – Sept. 2020
University Diploma of Technology (DUT), Computer Science <i>Paris Cité University, Paris, France</i>	Sept. 2017 – Sept. 2019

PUBLICATIONS

Boyu Zhu and Romain Giot. “Explaining Free-Text Keystroke Verification with OOD-aware SHAP.” *IEEE International Joint Conference on Biometrics (IJCB 2026)*. In press.

Boyu Zhu and Romain Giot. “Dataset Bias Detection for Biometric Verification with the Biometric Confusion Matrix: A Preliminary Experiment on Keystroke Dynamics.” *Explain'AI Workshop, Conférence francophone sur l'Extraction et la Gestion des Connaissances (EGC 2026)*. Oral presentation. [Full Paper]

Boyu Zhu and Romain Giot. “Biometric Confusion Matrix and Inter ZooPlot: Two Novel Visualizations for Biometric Verification Evaluation.” *IEEE International Joint Conference on Biometrics (IJCB 2025)*. Poster presentation. [Full Paper]

RESEARCH EXPERIENCE

PhD Student <i>LaBRI, University of Bordeaux</i>	Dec. 2024 – Present <i>Talence, France</i>
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- **Supervisor:** Romain GIOT, Associate Professor at University of Bordeaux
- **Research Topic:** Beyond Explainability and Evaluation Standards: A Better Understanding of Biometric Authentication
- Analyzed existing evaluation protocols and performance metrics in biometric authentication.
- Investigated global and local explainability methods for biometric authentication systems, with a focus on keystroke dynamics and face recognition.

Research Assistant Intern <i>IBISC Laboratory, Paris-Saclay University</i>	Feb. 2024 – Aug. 2024 <i>Évry, France</i>
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- **Supervisor:** Blaise HANCZAR, Full Professor at Paris-Saclay University
- **Research Topic:** Transformers in Precision Medicine
- Applied Transformer models for phenotype prediction from omics data.
- Investigated Transformer variants to optimize the trade-off between computational resources and training time, and analyzed their impact on prediction accuracy.

Research Assistant Intern <i>DAVID Laboratory, Paris-Saclay University</i>	Feb. 2023 – Aug. 2023 <i>Versailles, France</i>
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- **Supervisor:** Mustapha LEBBAH, Full Professor at Paris-Saclay University
- **Research Topic:** Attention-based Missing Value Imputation
- Proposed and investigated a Transformer-based method to improve prediction accuracy on datasets with missing values.
- Reviewed state-of-the-art literature and implemented models using Keras and TensorFlow.